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Measuring AI responsibility: A cross-country validation of a multidimensional framework

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ABSTRACT

As artificial intelligence (AI) continues to transform industries, ensuring AI responsibility has become critical for ethical governance. Despite the growing number of frameworks emphasizing transparency, accountability, and sustainability, a standardized measurement tool remains lacking. This study develops and validates a seven-dimensional AI Responsibility framework encompassing Privacy and Security, Transparency and Accountability, Impact on Employment, Sustainability, User-Centered Design, Social Impact, and Innovation and Adaptation. Using Exploratory Factor Analysis (EFA) and Confirmatory Factor Analysis (CFA), the study confirms the framework's construct validity and reliability. The results indicate strong model fit, with all constructs exceeding recommended thresholds for composite reliability (CR) and average variance extracted (AVE). The study contributes to AI ethics research by offering an empirically validated measurement instrument. Practically, the framework serves as a benchmarking tool for organizations and policymakers to assess AI governance strategies and regulatory compliance. As AI adoption continues to expand, this framework provides a structured approach to fostering trust, accountability, and responsible AI deployment.

1. Introduction

AI is rapidly transforming industries, societies, and the global economy, with increasing integration into decision-making processes across sectors such as healthcare, finance, manufacturing, and public services. Alongside these benefits, concerns regarding AI responsibility have emerged as a critical issue in AI governance and ethics (Smirnov & Lukyanov, 2019). The growing reliance on AI raises pressing questions related to accountability, transparency, privacy, sustainability, and social impact, highlighting the need for a structured framework to evaluate AI responsibility in organizational contexts. Despite ongoing efforts to establish ethical AI guidelines (Chakraborty et al., 2024), a standardized and empirically validated instrument for systematically measuring AI responsibility remains lacking.

Ensuring AI responsibility is essential for fostering trust, fairness, and accountability in AI systems. Ethical AI principles emphasize that AI should be developed and deployed in ways that align with human values and societal well-being (Dignum, 2020). As AI-driven decisions increasingly affect employment, privacy, security, and environmental sustainability, organizations are becoming more aware of the moral and legal responsibilities associated with AI adoption (Chatterjee et al., 2021; Krijger et al., 2023). However, AI responsibility remains a multidimensional and context-dependent construct, encompassing challenges such as algorithmic bias, limited explainability, data privacy risks, and workforce disruption (Amaka & Azeez, 2024; Olatunji et al., 2024). Although international bodies such as the OECD and the European Union have proposed AI governance frameworks, empirical validation of AI responsibility constructs remains limited, particularly in emerging market contexts where AI adoption is still evolving.

This study addresses this gap by developing and validating a measurement instrument to assess AI responsibility across organizational contexts. By empirically defining its key constructs, the study aims to provide a robust foundation for research and practice in AI ethics, corporate governance, and policy implementation. Drawing on prior literature, seven core dimensions of AI responsibility are identified: Privacy and Security, Transparency and Accountability, Impact on Employment, Sustainability, User-Centered Design, Social Impact, and Innovation and Adaptation (Amaka & Azeez, 2024; Chatterjee et al., 2021; Krijger et al., 2023; Luhana et al., 2023). Together, these dimensions offer a structured framework for evaluating the ethical, social, economic, and environmental implications of AI deployment, aligning with key international governance initiatives such as the European AI

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Act, the OECD AI Principles, and ISO AI standards.

Despite growing academic and regulatory attention to AI responsibility, no universally accepted measurement instrument currently captures its multidimensional nature. To address this gap, the present study develops and validates a quantitative AI Responsibility measurement scale using a rigorous psychometric approach. Specifically, the study addresses two research questions: (RQ1) What are the key constructs of AI responsibility? and (RQ2) How can the validity and reliability of an AI responsibility measurement instrument be established? The scale is empirically tested to ensure both theoretical coherence and statistical robustness.

This study makes two primary contributions. Theoretically, it consolidates fragmented discussions of AI responsibility into a coherent, empirically validated construct, thereby extending established Responsible AI frameworks (Floridi & Cowls, 2022; Jobin et al., 2019) by explicitly incorporating economic and environmental considerations. By operationalizing AI responsibility as a measurable multidimensional concept, the study strengthens the empirical foundations of AI ethics and governance research. Practically, it develops a validated AI Responsibility Measurement Instrument that offers organizations, policymakers, and AI developers a systematic tool for assessing, benchmarking, and improving responsible AI practices. More broadly, as AI continues to reshape organizational and societal processes, the ability to evaluate AI responsibility in a structured and evidence-based manner becomes increasingly important. The proposed framework provides a practical approach for examining responsible AI deployment across organizational contexts and national settings. By linking ethical principles with measurable indicators, this study contributes to bridging the gap between AI innovation and societal well-being, supporting the development of AI systems grounded in trust, accountability, and fairness.

2. Literature review

This literature review examines AI responsibility as an increasingly important construct in the context of ethical and sustainable AI deployment. Prior studies highlight that AI responsibility spans multiple domains, including transparency, accountability, and social impact. However, despite growing scholarly attention, literature lacks a standardized and empirically validated instrument capable of capturing these dimensions in an integrated manner. To address this gap, the present study develops and validates a multidimensional scale for measuring AI responsibility, providing a foundation for future empirical research and practical application.

2.1. AI responsibility

AI has become deeply embedded in economic and organizational systems, reshaping industries and decision-making processes worldwide (Smirnov & Lukyanov, 2019). As AI systems increasingly influence social, economic, and institutional outcomes, concerns regarding their responsible development and deployment have gained prominence. Prior research emphasizes that ethical AI should be guided by principles of transparency, accountability, and fairness, as these principles directly shape user trust and societal acceptance (Kaur, 2024; Sistla, 2024).

Ensuring AI responsibility extends beyond technical design and requires coordinated efforts among regulators, organizations, and technology developers. Collaborative governance mechanisms have been identified as essential for addressing ethical, legal, and social risks associated with AI adoption, including issues of accountability, bias, and misuse (Chakraborty et al., 2024; Krijger et al., 2023). Within this context, AI responsibility serves not only as a normative ideal but also as an operational concept that guides how AI systems are evaluated and governed in practice.

Conceptually, AI responsibility has been framed as a multidimensional construct encompassing the entire AI lifecycle. Dignum (2020) argues that responsibility must be embedded in the design, development, and use of AI systems, with transparency, accountability, and respect for human values as core principles. Complementing this view, Tigard (2021) distinguishes between moral, legal, and role-based responsibilities, highlighting the need to clarify responsibility attribution among different actors involved in AI systems. Together, these perspectives underscore that AI responsibility cannot be captured through a single principle or outcome but must be assessed across multiple normative dimensions.

Recent scholarship further emphasizes the importance of contextualizing AI responsibility within specific institutional, cultural, and application settings. Lee (2025) argue that universal ethical principles alone are insufficient to account for the socio-technical environments and domain-specific risks in which AI operates. Instead, responsible AI frameworks must balance shared ethical foundations with sensitivity to contextual variation across domains and national contexts. This perspective reinforces the need for measurement approaches that are theoretically grounded yet adaptable to diverse settings.

Building on these theoretical foundations, AI responsibility is conceptualized in this study as a multidimensional construct integrating ethical, legal, social, and organizational considerations. Consistent with prior frameworks, the proposed structure encompasses seven interrelated dimensions: Privacy and Security, Transparency and Accountability, Impact on Employment, Sustainability, User-Centered Design, Social Impact, and Innovation and Adaptation (Table 1). Together, these dimensions define the conceptual boundaries of AI responsibility and provide the basis for systematic measurement and empirical validation.

2.1.1. Privacy and security

Privacy and security serve as fundamental pillars of AI responsibility, playing a crucial role in fostering trust, transparency, and ethical AI deployment. According to Chatterjee et al. (2021), based on a survey of 324 respondents from Indian industries, concerns regarding Privacy and Security significantly influence stakeholders' trust and adoption of AI-integrated CRM systems. The study highlights risks associated with data breaches and the transparency of data management as key adoption barriers. Similarly, Krijger et al. (2023) conducted a survey across 1000 organizations from 20 industries and found that Privacy and Security are central to mitigating risks such as discrimination, reliability issues, and compliance challenges.

2.1.2. Transparency and accountability

Transparency and accountability are essential components of AI responsibility, ensuring trust, fairness, and ethical governance. Amaka and Azeez (2024) conducted surveys across multiple organizations, revealing that 78 % of respondents believe AI improves Transparency through clear audit trails, while 85 % emphasize Accountability in defining roles and responsibilities. Olatunji et al. (2024) collected data from AI developers, ethics experts, and AI users via surveys and case studies, showing that Explainable AI enhances Transparency by clarifying AI decisions, while strong governance frameworks improve Accountability, reducing bias. Ejjami (2024) surveyed executives and AI users in business contexts, highlighting that Transparency reduces misinformation and Accountability ensures clarity in roles and responsibilities.

2.1.3. Impact on employment

Research on Impact on Employment highlights its critical role in understanding AI Responsibility, focusing on workforce dynamics, psychological contracts, and skill adaptation. Luhana et al. (2023) surveyed 100 participants and found that while AI improves employee performance, it raises concerns about rising unemployment. Olaniyi et al. Olaniyi et al. (2024), using data from 482 professionals, revealed that AI creates new job opportunities but widens the skills gap, with 69 % of respondents emphasizing the need for urgent reskilling programs. Similarly, Braganza et al. (2021) analyzed 232 responses and discovered

Table 1
Constructs and Measurement Items of the AI Responsibility Scale.

Domain	Authors	Description	IOC	Corrected Item-Total Correlation
Privacy and Security	Chatterjee et al. (2021), Krijger et al. (2023)	AIR 1. Our organization has strong measures in place to protect user data from unauthorized access and data breaches.	1.00	0.541
		AIR 2. We ensure the integrity and confidentiality of user data through advanced protocols or technologies.	1.00	0.583
		AIR 3. We implement robust security protocols to prevent breaches and misuse of data.	1.00	0.596
		AIR 4. Our security measures are regularly reviewed and updated to address new threats.	1.00	0.553
Transparency and Accountability	Amaka Justina Obinna and Azeez Jason Kess-Momoh (2024), Olatunji Akinrinola et al. (2024)	AIR 5. The decision-making processes of our AI are documented and clearly communicated to users.	0.80	0.492
		AIR 6. Users are provided with clear and easily understandable explanations of AI decisions.	0.80	0.458
		AIR 7. We have established protocols to address failures of the AI system	1.00	0.626
		AIR 8. Responsible individuals are held accountable for issues arising from our AI systems.	0.80	0.535
Impact on Employment	Luhana et al. (2023), Olaniyi et al. (2024), Braganza et al. (2021)	AIR 9. Our organization conducts thorough analyses to assess potential job displacement caused by AI implementation.	1.00	0.591
		AIR 10. We actively address concerns related to job displacement by AI.	1.00	0.644
		AIR 11. We provide retraining and skill development opportunities for employees affected by AI.	1.00	0.588
		AIR 12. Our organization supports employees transitioning to new roles through effective programs.	1.00	0.558
Sustainability	Chang and Ke (2024), Khalid et al. (2024), Nosirov et al. (2024)	AIR 13. We make significant efforts to minimize the environmental impact of our AI systems.	1.00	0.455
		AIR 14. Our organization measures and reduces the carbon footprint of our AI technologies.	1.00	0.496
		AIR 15. Sustainable practices are integrated into the infrastructure of our AI systems.	1.00	0.561
		AIR 16. We use renewable resources in our AI operations to promote sustainability.	1.00	0.602
User-Centered Design	Padmasiri et al. (2023); Wang et al. (2023), Vincenzi et al. (2024)	AIR 17. Our AI systems are designed to be user-friendly and accessible.	1.00	0.506
		AIR 18. We use comprehensive usability testing methods to improve the AI interface.	0.80	0.504
		AIR 19. We have effective mechanisms for collecting and integrating user feedback into AI development.	1.00	0.502
		AIR 20. User feedback is reviewed and incorporated into regular AI system updates.	1.00	0.465
Social Impact	Hasas et al. (2024), Dumitrașcu et al. (2024), Cheng et al. (2021), Dumitrașcu et al. (2024),	AIR 21. Our organization actively engages with local and global communities in AI development.	1.00	0.513
		AIR 22. We have programs that support community involvement in AI projects.	1.00	0.528
		AIR 23. We ensure the ethical use of AI in social contexts through strict guidelines.	1.00	0.545
		AIR 24. Our organization adheres to ethical frameworks emphasizing fairness, responsibility, and transparency in AI.	1.00	0.577
Innovation and Adaptation	Russo (2024), Zhang and Suntrayuth (2024), Marshall et al. (2024)	AIR 25. Our organization is committed to continuously improving AI systems based on research and technological advancements.	1.00	0.455
		AIR 26. We stay updated on the latest developments in AI technology to ensure ongoing improvements.	0.80	0.523
		AIR 27. Our AI systems are designed to be flexible and adaptable to changing societal needs.	1.00	0.564
		AIR 28. We have strategies to ensure our AI systems remain relevant and effective in dynamic environments.	1.00	0.596

that AI adoption enhances employee engagement and trust, but disrupts established psychological contracts.

2.1.4. *Sustainability*

Sustainability emerges as a critical subdomain of AI Responsibility, focusing on environmental stewardship, social equity, and long-term economic viability. Chang and Ke (2024) analyzed 75 studies, integrating Sustainability, CSR (Corporate Social Responsibility), ESG (Environmental, Social, and Governance), and SDGs (Sustainable Development Goals) into an AI-enabled analytics framework. They emphasized AI's potential to enhance transparency and accountability in organizational sustainability practices. Similarly, Khalid et al. (2024) surveyed 428 senior managers and found that AI-driven risk management significantly supports sustainable decision-making by optimizing resource allocation and mitigating environmental risks. Lastly, Nosirov et al. (2024) combined survey data and case studies to demonstrate AI's effectiveness in improving resource efficiency, reducing waste, and achieving sustainability targets, despite challenges like high energy consumption.

Recent work by Al-Maaitah et al. (2025) further emphasizes that AI-driven sustainability initiatives should be evaluated not only in terms of efficiency gains, but also through the lens of ethical governance and long-term societal accountability. Her study highlights that responsible AI practices strengthen sustainable decision-making by aligning technological innovation with environmental and social responsibility frameworks. This perspective reinforces the conceptualization of Sustainability as a core dimension of AI Responsibility, extending it beyond environmental performance to include ethical oversight and

institutional accountability.

2.1.5. User-Centered design

User-Centered Design plays a pivotal role in shaping AI Responsibility by prioritizing user needs, transparency, and fairness throughout the AI development lifecycle. Wang et al., Wang et al. (2023) analyzed user experience practices in responsible AI systems and found that UCD methodologies address challenges related to fairness, accountability, and transparency, ensuring AI systems align with user expectations and ethical guidelines. Vincenzi et al. (2024) conducted a survey with 1121 respondents in the UK, revealing that lay users express significant concerns about AI fairness and transparency and emphasize the importance of user involvement across all AI development stages. Similarly, Padmasiri et al. (2023) surveyed 150 user experience professionals in Sri Lanka and demonstrated that AI integration enhances user experience processes, enabling more personalized and efficient user experiences while addressing design challenges.

2.1.6. Social impact

In the context of rapidly advancing and generative AI systems, social impact is no longer a passive outcome of technology adoption but a dynamic ethical process through which societies negotiate fairness, inclusion, and power relations in human–AI interaction.

Accordingly, this dimension is conceptualized not merely in terms of social outcomes, but as a core mechanism through which AI responsibility is socially constituted and contested. This perspective shifts social impact from a descriptive outcome variable to a constitutive ethical dimension of AI responsibility.

Social Impact plays a crucial role in shaping AI Responsibility, highlighting AI's potential to address societal challenges while mitigating risks associated with fairness, equity, and accessibility. Hasas et al. (2024) surveyed 120 participants and found that AI contributes significantly to healthcare, education, and environmental sustainability, improving societal well-being. Similarly, Dumitrașcu et al. (2024) surveyed 309 healthcare professionals and revealed positive perceptions of AI's role in enhancing healthcare outcomes, despite limited technical knowledge among professionals. Furthermore, Cheng et al. (2021) emphasized the importance of socially responsible AI algorithms to reduce algorithmic bias and promote fairness.

Taken together, these perspectives position Social Impact and its close interaction with Innovation and Adaptation not merely as an outcome of AI deployment, but as a dynamic ethical dimension that reflects how societies respond to the disruptive and generative capacities of advanced AI systems.

2.1.7. Innovation and adaptation

In environments shaped by generative AI and emerging AGI trajectories, innovation and adaptation must be understood not only as technological progress but as an ethical and institutional capacity to govern continuous socio-technical change. This dimension therefore captures how organizations and societies adjust governance, norms, and practices in response to the evolving risks and opportunities created by advanced AI systems. Accordingly, innovation and adaptation are treated as governance capacities rather than merely technical capabilities.

Innovation and Adaptation are critical to AI Responsibility because they ensure that AI technologies remain aligned with organizational goals and societal needs. Empirical evidence shows that compatibility with existing workflows (Russo, 2024), organizational agility and ambidextrous leadership (Zhang & Suntrayuth, 2024), and innovation-driven business models (Marshall et al., 2024) enable the responsible design, deployment, and continuous improvement of AI systems.

Beyond these organizational factors, generative AI introduces profound socio-technical changes in knowledge creation, education, and human–AI interaction (Yeralan & Lee, 2023). These changes require ongoing adaptation of governance mechanisms, ethical norms, and institutional practices. Accordingly, within the AI responsibility framework, Innovation and Adaptation represent not only technological advancement but also the ethical and institutional capacity to respond responsibly to the disruptive and evolving nature of advanced AI systems.

2.1.8. Contextual ethical frameworks for AI responsibility

Recent research emphasizes that AI Responsibility must be grounded in domain-specific and context-sensitive ethical frameworks, particularly in sustainability-oriented and ESG-related applications. Areiqat and Jaber (2025) and Al-Maaitah et al. (2025) show that responsible AI adoption requires strong governance structures to ensure transparency, accountability, and long-term value creation while mitigating risks such as algorithmic bias and opacity.

At the same time, foundational ethical principles including transparency, accountability, and sustainability remain broadly shared across contexts, enabling the development of common measurement frameworks (Floridi & Cowls, 2022; Jobin et al., 2019). Accordingly, the present study proposes a seven-dimensional AI Responsibility framework that captures both universal ethical foundations and their contextual manifestations across national and institutional settings.

2.2. Developing tools for AI responsibility

The development of measurement tools for AI responsibility is increasingly necessary to support ethical standards, accountability, and transparent AI deployment across organizational and societal contexts.

2.2.1. Theoretical foundations for tool development

Responsibility Gap Theory provides a key theoretical rationale for developing tools to assess AI responsibility (Jonas, 1984; Matthias, 2004). As AI systems operate with increasing autonomy and complexity, traditional mechanisms for assigning moral and legal responsibility become insufficient. Santoni De Sio and Mecacci (2021) emphasize the need to address gaps in culpability and accountability, while Coeckelbergh (2020) highlights the importance of explainability and relational justification in sustaining responsible human–AI interactions. Together, these perspectives underscore the need for systematic measurement tools capable of evaluating AI responsibility across ethical, social, and technical dimensions.

2.2.2. Component of effective assessment tools

To develop a scale for measuring AI responsibility, it is essential to include dimensions such as accuracy and actionable insight. The Likert Scale, originally introduced by Likert (1932), is widely recognized as an effective method for measuring self-reported competencies. Likert scales demonstrate high accuracy across multiple domains, including seven identified subdomains. To ensure the reliability and validity of this scale, both EFA and CFA should be employed. These statistical methods help validate the theoretical framework and confirm the robustness of the assessment tool.

2.2.3. Research gap for AI responsibility

The existing literature lacks a comprehensive and standardized measurement scale for assessing AI responsibility across multiple domains. Current studies predominantly focus on isolated aspects, such as transparency or the impact on employment, without offering an integrated approach. To address this gap, this study aims to develop a robust and multidimensional measurement tool covering key domains, including privacy and security, transparency and accountability, impact on employment, sustainability, and others. By collaborating with academic experts specializing in AI, this study ensures the proposed scale is both reliable and contextually relevant, offering a valuable contribution to the field of AI responsibility research.

3. Methodology

This study employed a structured, multi-phase design to develop and validate a measurement instrument for AI responsibility. The methodology comprised three stages: scale development, sampling and data collection, and data analysis. Scale development began with a focused review of AI ethics and governance literature to identify key dimensions of AI responsibility. Through thematic synthesis, seven core dimensions were specified: Privacy and Security, Transparency and Accountability, Impact on Employment, Sustainability, User-Centered Design, Social Impact, and Innovation and Adaptation. A preliminary instrument consisting of 28 items was developed to operationalize these dimensions.

Data were collected using a two-stage sampling design consistent with psychometric validation practices. The required sample size was determined based on a minimum ratio of 15 respondents per item, yielding a target sample of 840 participants. Faculty members from business schools were selected due to their familiarity with AI applications in organizational and educational contexts. In the first stage, EFA was conducted using responses from 420 faculty members at universities in Guangxi Zhuang Autonomous Region, China, to identify the underlying factor structure and refine measurement items. In the second stage, CFA was performed on an independent sample of 420 faculty members from nine Rajamangala Universities of Technology in Thailand to validate the factor structure and assess construct validity and model fit.

Data analysis followed a two-step procedure. EFA was used to explore latent dimensions and eliminate weak or redundant items. CFA was subsequently applied to evaluate the measurement model, including composite reliability, average variance extracted, and discriminant validity. Model fit was assessed using standard goodness-of-fit indices. To examine the cross-country equivalence of the proposed measurement model, multi-group confirmatory factor analysis (MG-CFA) was conducted across the Chinese and Thai samples. Configural invariance was first assessed to determine whether the same factor structure was applicable across groups, followed by metric invariance testing by constraining factor loadings to be equal. Model comparisons were evaluated based on changes in CFI and RMSEA in accordance with established guidelines. For the measurement invariance analysis, the finalized measurement model established through single-group CFA was used as the baseline across both samples. No additional post-hoc model modifications were planned or introduced during the invariance testing in order to preserve cross-group comparability.

4. Results

This section presents the empirical findings of the study, structured across three sequential phases of scale development and validation. Phase 1 reports the instrument development and content validation process. Phase 2 and Phase 3 present the results of exploratory and confirmatory factor analyses conducted to establish the construct validity and reliability of the AI Responsibility scale.

Phase 1. Instrument Development and Content Validation.

The development of the AI Responsibility measurement instrument followed established scale construction principles, with particular emphasis on content validity and reliability. Based on an extensive synthesis of the AI governance and responsible AI literature, seven core domains were identified: Privacy and Security, Transparency and Accountability, Impact on Employment, Sustainability, User-Centered Design, Social Impact, and Innovation and Adaptation. These domains formed the conceptual foundation of the instrument. For each domain, key constructs and operational definitions were specified, drawing on prior frameworks and empirical studies to ensure theoretical coherence. Questionnaire items were then formulated to reflect the defining characteristics of each domain. The initial instrument comprised 28 items, systematically distributed across the seven dimensions.

Content validity was assessed by an expert panel consisting of five specialists with expertise in information technology, organizational behavior, and research methodology. The experts evaluated each item for clarity, relevance, and alignment with its intended construct. Item–Objective Congruence (IOC) values indicated a high level of agreement, with all items meeting accepted content validity criteria. Internal consistency was subsequently examined using corrected item–total correlations and Cronbach's alpha. Corrected item–total correlations were within acceptable ranges, and the overall Cronbach's alpha coefficient was 0.927, indicating strong internal reliability of the scale. Together, these results confirm the adequacy of the instrument for subsequent exploratory and confirmatory factor analyses. Table 1 summarizes the constructs, representative measurement items, and supporting literature of the AI Responsibility scale.

Phase 2. Exploratory Factor Analysis.

This phase examined the underlying factor structure of the AI Responsibility scale and refined the measurement instrument prior to confirmatory validation. EFA was employed to assess whether the proposed items clustered into coherent and theoretically meaningful constructs.

The EFA sample consisted of 387 Chinese lecturers from 18 public universities in Guangxi Zhuang Autonomous Region, China (response rate 90 %). The sample showed a balanced gender distribution (50.96 % female, 49.04 % male), diverse educational levels (44.81 % bachelor's, 35.00 % master's, and 20.19 % doctoral), and a wide range of professional experience, with 50.58 % having 0–5 years and the remainder distributed across higher experience levels, indicating a heterogeneous and appropriate sample for exploratory factor analysis.

Prior to analysis, the dataset was screened for completeness and quality. Only fully completed questionnaires were retained, and no evidence of inattentive or patterned responding was identified. Preliminary diagnostics indicated no multicollinearity concerns, as inter-item correlations remained below commonly accepted thresholds.

Sampling adequacy was evaluated using the Kaiser–Meyer–Olkin (KMO) measure and Bartlett's test of sphericity. The KMO value of 0.920 indicated excellent sampling adequacy, while Bartlett's test was statistically significant ($\chi^2 = 4974.833$, $p < 0.001$), confirming the suitability of the data for factor analysis. Factors were extracted using principal component analysis with varimax rotation. Applying the eigenvalue-greater-than-one criterion resulted in a seven-factor solution, accounting for 65.489 % of the total variance.

Item communalities ranged from 0.532 to 0.830, indicating that the extracted factors adequately explained item variance. As shown in Table 2, all items loaded strongly on their intended factors, with factor loadings ranging from 0.599 to 0.859 and no problematic cross-loadings observed. The resulting factor structure was fully consistent with the theoretically proposed dimensions of AI responsibility.

Table 2 presents the EFA results, confirming seven distinct factors: Privacy and Security, Social Impact, Innovation and Adaptation, Impact on Employment, Transparency and Accountability, Sustainability, and User-Centered Design. At the conclusion of Phase 2, the first research question "What are the key constructs of AI responsibility?" was addressed. The analysis empirically validated these seven constructs, providing a stable measurement structure for subsequent confirmatory factor analysis.

Phase 3. Major study for confirming the scale's construct validity.

Following the establishment of the factor structure in Phase 2, CFA was conducted to validate the measurement model and assess construct validity, reliability, and overall model fit. An independent sample of university faculty members from Thailand was used to confirm the robustness of the proposed seven-factor structure.

The CFA sample consisted of 310 Thai lecturers (response rate 73 %), with a gender distribution of 58.4 % male and 41.6 % female. The sample showed high educational attainment, with 50.6 % holding a

Table 2
Exploratory Factor Analysis Results of the AI Responsibility Scale.

Item	Common Factor	Component						
		I	II	III	IV	V	VI	VII
AIR 1	0.691	0.777						
AIR 2	0.830	0.859						
AIR 3	0.761	0.807						
AIR 4	0.532	0.611						
AIR 5	0.630					0.662		
AIR 6	0.605					0.726		
AIR 7	0.668					0.668		
AIR 8	0.687					0.741		
AIR 9	0.647				0.666			
AIR 10	0.651				0.643			
AIR 11	0.627				0.676			
AIR 12	0.693				0.694			
AIR 13	0.606						0.628	
AIR 14	0.622						0.726	
AIR 15	0.747						0.788	
AIR 16	0.631						0.599	
AIR 17	0.571							0.649
AIR 18	0.694							0.762
AIR 19	0.644							0.704
AIR 20	0.605							0.636
AIR 21	0.618		0.683					
AIR 22	0.693		0.737					
AIR 23	0.718		0.766					
AIR 24	0.628		0.645					
AIR 25	0.588			0.705				
AIR 26	0.661			0.707				
AIR 27	0.702			0.749				
AIR 28	0.587			0.601				

Note: The extraction method used was principal factor analysis and the rotation method was.

master's degree, 44.8 % a doctoral degree, and 4.5 % a bachelor's degree, as well as a wide range of professional experience, ranging from early-career academics to those with >25 years of service, indicating a heterogeneous and appropriate sample for confirmatory factor analysis.

The CFA model demonstrated an excellent fit to the data (χ^2 = 328.907, df = 297, p = 0.098; χ^2/df = 1.107; CFI = 0.995; TLI = 0.994; GFI = 0.928; AGFI = 0.901; RMSEA = 0.019), indicating strong consistency between the empirical data and the hypothesized measurement structure (Fig. 1). All standardized factor loadings were substantial and statistically significant, ranging from 0.71 to 0.92. The first-order CFA supported the validity of the seven dimensions, while a second-order CFA confirmed that they collectively represented AI responsibility as a higher-order construct (standardized loadings = 0.76–0.99).

Construct reliability and convergent validity were further supported by composite reliability (CR) and average variance extracted (AVE) values. CR values exceeded recommended thresholds for all first-order constructs (ranging from 0.83 to 0.92) and for the second-order construct (CR = 0.95). AVE values ranged from 0.56 to 0.75, indicating adequate convergent validity (Table 3).

To examine cross-country equivalence, measurement invariance of the AI Responsibility scale across the Chinese and Thai samples was tested using multi-group CFA. A sequence of increasingly constrained models was estimated. The configural model demonstrated an excellent fit, indicating a common factor structure across groups. Metric invariance was supported, as constraining factor loadings resulted in negligible changes in model fit (ΔCFI = −0.003; ΔRMSEA = +0.002). Scalar invariance was subsequently established by constraining item intercepts, with changes in fit indices remaining within accepted thresholds (ΔCFI = +0.009; ΔRMSEA = +0.007). No post-hoc model modifications were introduced during invariance testing, ensuring strict comparability across groups (Table 5). Overall, the results support cross-country equivalence of the AI Responsibility scale across the Chinese and Thai samples.

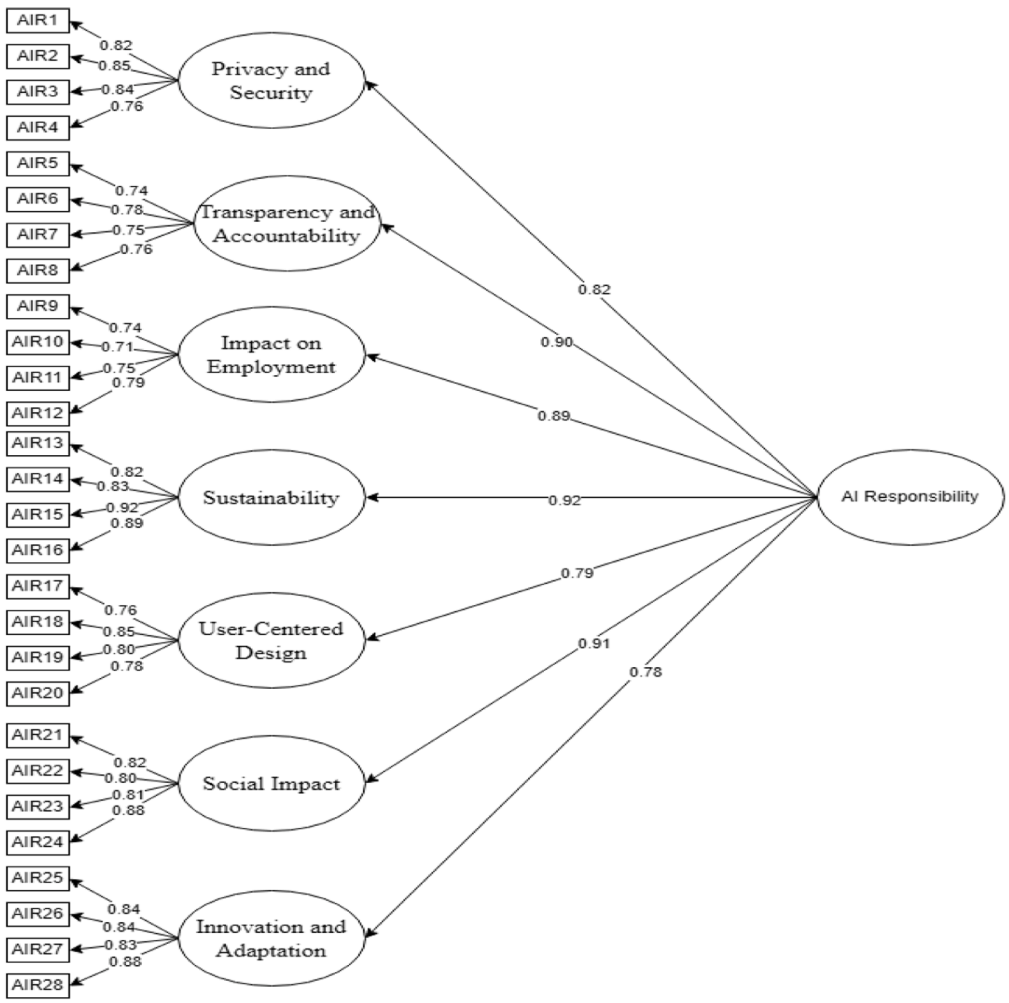
At the conclusion of Phase 3, the second research question "How can the validity and reliability of an AI responsibility measurement instrument be established?" was addressed. The results confirm that the AI Responsibility scale demonstrates strong construct validity, reliability, and cross-country measurement equivalence, supporting its use in comparative and cross-national research contexts.

5. Discussion

This study aimed to identify the key constructs of AI responsibility (RQ1) and to establish the validity and reliability of its measurement instrument (RQ2). The findings make two primary contributions. First, the study consolidates and refines the theoretical foundations of AI responsibility by articulating its core dimensions. Second, it develops and empirically validates a comprehensive measurement instrument that enables AI responsibility to be assessed in a systematic and replicable manner.

The results confirm that AI responsibility is a multidimensional construct comprising seven core dimensions: Privacy and Security, Transparency and Accountability, Impact on Employment, Sustainability, User-Centered Design, Social Impact, and Innovation and Adaptation. These dimensions are consistent with established ethical AI principles (Floridi & Cowls, 2022; Jobin et al., 2019) and collectively capture how AI systems should be designed, governed, and deployed to achieve socially beneficial outcomes.

Privacy and Security reflects the protection of personal data and system integrity, while Transparency and Accountability addresses the explainability of AI systems and responsibility for AI-driven decisions (Chatterjee et al., 2021; Olatunji et al., 2024). Impact on Employment captures how organizations manage workforce displacement and reskilling in response to automation (Autor, 2015; Luhana et al., 2023). Sustainability reflects the environmental implications of AI deployment, while User-Centered Design emphasizes usability, accessibility, and alignment with human needs (Sitnikov, 2013; Wang et al., 2023). Social Impact addresses the broader societal consequences of AI, particularly for vulnerable groups (Cheng et al., 2021; Hasas et al., 2024). Innovation and Adaptation reflects the capacity of organizations to continuously



Chi-square = 328.907, df = 297, p = 0.098, Chi-square /DF = 1.107, GFI = 0.928,

AGFI= 0.901, PGFI = 0.679, RMSEA = 0.019

Fig. 1. AI Responsibility Structural Model.

evolve AI systems in response to technological and societal change (Rogers et al., 2019).

To address RQ2, the study employed EFA, CFA, and reliability and validity assessments. EFA confirmed a clear seven-factor structure, with all items loading strongly on their intended constructs and showing no problematic cross-loadings, indicating that the proposed dimensions are empirically distinct and conceptually coherent. CFA further demonstrated excellent model fit, with strong incremental fit indices and minimal approximation error, confirming that the hypothesized structure closely represents the observed data. Reliability and validity analyses also met established thresholds, with high composite reliability, adequate convergent validity, and clear discriminant validity across all constructs.

Beyond establishing internal validity, the present study extends prior work by formally examining the cross-country equivalence of the AI Responsibility measurement framework. Multi-group confirmatory factor analysis provides clear evidence of configural, metric, and scalar invariance across the Chinese and Thai samples, indicating that respondents in both national contexts conceptualize AI responsibility through the same seven-dimensional structure and that the measurement items operate equivalently in terms of factor loadings, construct relationships, and intercepts. The establishment of metric and scalar invariance enables meaningful cross-country comparisons and provides strong empirical support for the robustness and cross-national applicability of the proposed AI Responsibility instrument.

Taken together, these findings confirm that the AI Responsibility instrument is valid, reliable, and suitable for use in comparative and cross-national research. The framework provides a robust foundation for future empirical studies and policy-oriented assessments of ethical and responsible AI.

Beyond its measurement function, the proposed AI Responsibility framework offers a theoretically grounded structure for sector-specific application. In smart city contexts, dimensions such as Transparency and Accountability, Privacy and Security, and Social Impact are particularly important for ethical data governance, public trust, and inclusive urban services. In educational settings, Innovation and Adaptation, User-Centered Design, and Social Impact become central for addressing challenges associated with generative AI, academic integrity, and evolving human–AI interaction. These illustrative mappings demonstrate how the framework can be conceptually applied across domains to guide empirical implementation and policy design.

Table 3
Construct Validity of the First and Second Order CFA Results.

Latent Variables/Questions	First order				Second order			
	Factor Loading	R^2	CR	AVE	Factor Loading	R^2	CR	AVE
AI Responsibility							0.95	0.74
Privacy and Security			0.89	0.67	0.76	0.58		
AIR 1	0.82	0.67						
AIR 2	0.84	0.72						
AIR 3	0.85	0.71						
AIR 4	0.76	0.58						
Transparency and Accountability			0.84	0.57	0.92	0.85		
AIR 5	0.74	0.55						
AIR 6	0.78	0.61						
AIR 7	0.75	0.56						
AIR 8	0.76	0.58						
Impact on Employment			0.83	0.56	0.99	0.98		
AIR 9	0.74	0.55						
AIR 10	0.71	0.50						
AIR 11	0.75	0.56						
AIR 12	0.79	0.62						
Sustainability			0.92	0.75	0.90	0.81		
AIR 13	0.82	0.67						
AIR 14	0.83	0.69						
AIR 15	0.92	0.85						
AIR 16	0.89	0.79						
User-Centered Design			0.87	0.63	0.80	0.64		
AIR 17	0.76	0.58						
AIR 18	0.85	0.72						
AIR 19	0.80	0.64						
AIR 20	0.78	0.61						
Social Impact			0.88	0.65	0.87	0.76		
AIR 21	0.82	0.67						
AIR 22	0.80	0.64						
AIR 23	0.81	0.66						
AIR 24	0.80	0.68						
Innovation and Adaptation			0.90	0.71	0.79	0.62		
AIR 25	0.84	0.71						
AIR 26	0.84	0.71						
AIR 27	0.83	0.69						
AIR 28	0.88	0.77						

Discriminant validity was assessed using the Fornell–Larcker criterion. As reported in Table 4, the square root of AVE for each construct exceeded the corresponding inter-construct correlations, providing evidence of satisfactory discriminant validity.

Table 4
Discriminant Validity Analysis.

Variable	1.	2.	3.	4.	5.	6.	7.
Privacy and Security	(0.82)						
Transparency and Accountability	0.73**	(0.76)					
Impact on Employment	0.64**	0.75**	(0.75)				
User-Centered Design	0.64**	0.72**	0.75**	(0.87)			
Social Impact and Innovation	0.55**	0.61**	0.69**	0.63**	(0.80)		
Social Impact and Innovation	0.65**	0.66**	0.67**	0.74**	0.61**	(0.81)	
Adaptation	0.52**	0.61**	0.63**	0.65**	0.66**	0.77**	(0.86)

Note:.
** indicates 0.01 significance level, values in parentheses represent the square root of AVE.

Table 5
Measurement Invariance Testing Across Chinese and Thai Samples.

Model	χ^2 (CMIN)	df	CFI	RMSEA	ΔCFI	ΔRMSEA
Configural (Unconstrained)	785.149	562	0.980	0.024	—	—
Metric (Measurement Weights)	839.068	583	0.977	0.026	−0.003	+0.002
Structural Weights	854.695	589	0.976	0.026	−0.001	0.000
Structural Covariances	918.713	590	0.971	0.029	−0.005	+0.003
Structural Residuals	951.376	597	0.969	0.030	−0.002	+0.001
Measurement Residuals	1343.992	682	0.941	0.038	−0.028	+0.008
Scalar Invariance	369.853	281	0.986	0.033	+ 0.009	+ 0.007

Note: ΔCFI and ΔRMSEA are calculated relative to the metric invariance model for scalar invariance and relative to the immediately preceding, less constrained model for other steps. Changes in CFI $\leq$ 0.010 and RMSEA $\leq$ 0.015 indicate acceptable invariance.

6. Theoretical implications

The findings of this study contribute to the theoretical advancement of AI responsibility by offering an empirically validated and multidimensional framework that integrates core principles from AI ethics, governance, and responsible AI scholarship. The seven dimensions identified Privacy and Security, Transparency and Accountability, Impact on Employment, Sustainability, User-Centered Design, Social Impact, and Innovation and Adaptation provide a structured and measurable conceptualization of AI responsibility that extends existing theoretical perspectives.

First, the results reinforce and operationalize the Responsible AI frameworks proposed by Floridi and Cowls (2022) and Jobin et al. (2019), which emphasize transparency, accountability, and societal benefit as foundational principles of ethical AI. The dimensions of Privacy and Security, Transparency and Accountability, and Social Impact directly align with these principles, underscoring the importance of safeguarding users, ensuring explainability, and mitigating social harm. Importantly, this study extends prior frameworks by explicitly incorporating Impact on Employment and Sustainability, thereby broadening the theoretical scope of AI responsibility beyond ethical design considerations to include economic and environmental consequences of AI adoption.

Second, drawing on the construct validity criteria proposed by Fornell and Larcker (1981), this study provides empirical evidence that AI responsibility constitutes a distinct and measurable latent construct. The application of EFA and CFA confirms the stability and coherence of the seven-factor structure, supporting the operationalization of AI responsibility for both academic inquiry and applied research. The strong composite reliability and average variance extracted values further demonstrate that the identified dimensions possess robust theoretical and empirical foundations across organizational and regulatory contexts.

In addition, the inclusion of Impact on Employment as a core dimension extends existing discussions on automation and labor markets, particularly those advanced by Autor (2015). While much of the prior literature frames automation primarily as an economic or technological phenomenon, the present findings highlight employment-related outcomes as an ethical and governance concern. This perspective suggests that AI responsibility encompasses not only transparency and fairness in algorithmic systems, but also organizational accountability for workforce transitions, reskilling, and social sustainability.

Finally, the validated framework contributes to ongoing debates surrounding global AI governance initiatives, including the European AI Act and the OECD AI Principles, which emphasize risk management, accountability, and transparency in AI deployment. By providing an empirically grounded measurement structure, this study offers a theoretical basis for translating regulatory principles into assessable constructs. The alignment of key dimensions such as Privacy and Security, Transparency and Accountability, and Social Impact with international regulatory agendas reinforces the need for standardized tools that support consistent and responsible AI governance across jurisdictions.

7. Practical implications

The findings of this study offer important practical implications for policymakers, business leaders, AI developers, and organizations seeking to implement responsible AI practices. By empirically validating a seven-dimensional AI responsibility framework, this study provides a structured and actionable approach for assessing, monitoring, and improving AI governance in line with ethical, legal, and societal expectations.

First, for policymakers and regulatory authorities, the validated measurement instrument offers a practical foundation for operationalizing AI governance initiatives such as the European AI Act, the OECD AI Principles, and emerging ISO AI standards. Core dimensions including Transparency and Accountability, Privacy and Security, and Social Impact directly correspond to regulatory requirements related to explainability, risk management, and ethical AI deployment. The framework enables regulators to develop standardized assessment criteria, facilitating consistent evaluation of AI systems across industries and jurisdictions. Moreover, the inclusion of Sustainability highlights the growing importance of assessing the environmental impacts of AI, supporting the integration of Green AI considerations into future regulatory and policy frameworks.

Second, for business leaders and AI-driven organizations, the findings underscore the strategic role of AI responsibility within corporate governance and decision-making processes. Organizations can apply the framework to systematically evaluate their AI systems, ensuring compliance with best practices in data protection, ethical deployment, and responsible workforce management. In particular, the Impact on Employment dimension emphasizes the need for proactive workforce strategies, such as reskilling initiatives, job transition planning, and responsible human–AI collaboration. Rather than focusing solely on efficiency gains, organizations are encouraged to address the social implications of AI adoption, thereby strengthening corporate social responsibility (CSR) commitments and enhancing stakeholder trust.

Third, for AI developers and system designers, the study highlights the importance of embedding User-Centered Design principles throughout the AI development lifecycle. Developers are encouraged to prioritize usability, accessibility, and fairness, while minimizing algorithmic bias and improving transparency. The validated framework further suggests that explainability mechanisms, human oversight, and adaptive learning capabilities should be integral components of AI systems. In addition, the Innovation and Adaptation dimension underscores the need for continuous model evaluation and iterative updates, ensuring that AI technologies remain responsive to evolving societal expectations and ethical standards.

Fourth, for organizations implementing AI technologies, the framework serves as a benchmarking tool for evaluating AI responsibility practices against industry norms. By applying the model, organizations can identify gaps in transparency, accountability, sustainability, and social impact, enabling targeted improvements in AI governance. The framework may also be incorporated into ESG (Environmental, Social, and Governance) reporting, positioning AI governance as a core element of broader corporate sustainability and accountability strategies.

Finally, for academic research and future studies, this validated instrument provides a reliable foundation for cross-industry and cross-country comparisons of AI responsibility. Researchers may use the framework to examine how AI responsibility is operationalized across sectors and regulatory environments, as well as to evaluate the effectiveness of different AI governance approaches. The inclusion of dimensions such as Privacy and Security, Transparency and Accountability, and Social Impact makes the instrument particularly well suited for research in highly regulated domains, including healthcare, finance, and defense.

8. Conclusion

This study contributes to the growing discourse on AI responsibility by developing and validating a comprehensive measurement framework that captures the multidimensional nature of responsible AI deployment. By identifying and empirically validating seven key dimensions Privacy and Security, Transparency and Accountability, Impact on Employment, Sustainability, User-Centered Design, Social Impact, and Innovation and Adaptation, this research provides a structured and measurable approach to assessing AI responsibility. The findings reinforce existing AI governance principles, such as the Responsible AI Framework (Floridi & Cowls, 2022; Jobin et al., 2019), the European AI Act, and the OECD AI Principles, while extending their applicability by integrating economic and environmental considerations into the AI responsibility

framework.

The rigorous validation process, incorporating EFA and CFA, confirms the theoretical soundness and empirical robustness of the proposed measurement instrument. The results demonstrate strong model fit indices, with high reliability and validity scores, establishing AI responsibility as a quantifiable construct that can be applied in both academic research and industry practice. Organizations, policymakers, and AI developers can leverage this instrument to evaluate AI governance strategies, benchmark AI responsibility metrics, and implement responsible AI practices that align with ethical and regulatory standards.

By systematically measuring AI responsibility, this study addresses a critical gap in AI ethics research, providing a foundational tool for assessing and improving the ethical, social, economic, and environmental implications of AI adoption. As AI technologies continue to evolve and integrate into various sectors, ensuring accountability, fairness, and sustainability will be paramount. This study serves as a stepping stone for future AI governance research, encouraging further exploration of AI responsibility across industries, regions, and regulatory environments.

9. Limitations and future research

Despite its contributions, this study is subject to several limitations that should be acknowledged and addressed in future research. First, empirical validation was conducted within specific organizational and regulatory contexts, which may constrain the generalizability of the findings to other regions or industries. Although the measurement instrument demonstrated strong psychometric properties, its applicability to different AI governance environments particularly those characterized by rapidly evolving regulatory regimes warrants further examination. Future studies should therefore extend validation efforts across industries and countries, including sectors such as smart cities, education, and defense, where AI adoption raises distinct ethical, operational, and regulatory challenges.

Second, the study employed samples from two national contexts, with data from China used for exploratory factor analysis and data from Thailand used for confirmatory factor analysis. This design was intended to enhance external validity by testing the scale across different settings. However, the use of distinct populations may also introduce cultural and institutional influences that cannot be fully disentangled from measurement effects. Although satisfactory model fit was achieved, explicit measurement invariance testing was not the primary focus of this study. Future research should therefore employ multi-group confirmatory factor analysis to formally establish configural, metric, and scalar invariance and further strengthen the cross-cultural applicability of the instrument. Third, the study relied on self-reported survey data, which may be subject to social desirability bias, particularly in relation to ethical and governance-related constructs. Respondents may overstate their organization's commitment to responsible AI practices. To mitigate this limitation, future research could incorporate complementary methodologies such as interviews, case studies, or document analysis, as well as longitudinal designs to capture how AI responsibility evolves over time in response to regulatory and organizational changes.

Fourth, while the proposed framework identifies seven core dimensions of AI responsibility, ongoing technological developments may give rise to new ethical and governance challenges. Issues such as algorithmic fairness, explainability, human–AI collaboration, and the governance of generative and autonomous AI systems may necessitate further refinement or expansion of the framework. Future studies should examine whether additional constructs are required to reflect emerging risks and evolving societal expectations surrounding advanced AI technologies. Fifth, the practical implementation of AI responsibility metrics remains an important avenue for future research. Empirical investigations could examine how organizations integrate AI responsibility into strategic decision-making, corporate governance structures, and risk management processes. In addition, the intersection of AI responsibility with emerging technologies such as blockchain, federated learning, and quantum computing represents a promising area for further inquiry.

Sixth, some items in the instrument were formulated in aspirational rather than strictly behavioral terms. While such wording captures normative expectations and organizational intentions, it may not fully reflect actual practices. Future research should consider refining item phrasing to more clearly distinguish between aspirational commitments and observable behaviors, thereby enhancing the interpretability of scale scores. Finally, although the instrument demonstrated strong reliability and validity, external benchmarks for interpreting scale scores were not established. As a result, distinctions between "high" and "low" levels of AI responsibility remain relative rather than absolute. Future studies should link the scale to external criteria, such as organizational performance indicators, regulatory compliance measures, or third-party assessments, to strengthen its practical applicability.

By addressing these limitations, future research can further refine and extend the proposed AI responsibility framework. Such efforts will be critical for supporting evidence-based AI governance, informing policy development, and guiding organizational best practices, ultimately fostering AI systems that are not only innovative and efficient but also ethical, transparent, and socially accountable.

CRediT authorship contribution statement

Nutt Jaturat: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Khahan Na-Nan:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Bowei Hu:** Writing – original draft, Methodology, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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